

Software Design Specification: Online Appointment System

System Introduction & Requirements Analysis Baseline

DOCUMENT DOMAIN	SYSTEM NAME	DOCUMENT PHASE	STATUS
Software Engineering	Online Appointment System	Introduction / SRS	Draft / Complete

1. Background

In traditional service-oriented environments—such as healthcare clinics, academic advising centers, auto service hubs, and professional consulting offices—appointment scheduling relies heavily on manual coordination. Clients frequently experience long wait times, missed calls during peak operational hours, and scheduling conflicts due to non-centralized physical logbooks or disconnected communication channels.

With the widespread adoption of modern web-based solutions, transitioning to an automated, digital appointment management ecosystem streamlines administrative workflows, minimizes operational overhead, and enables 24/7 self-service scheduling for end users.

2. Problem Statement & Need of Project

Manual scheduling systems present several operational bottlenecks that negatively impact organizational efficiency and customer satisfaction:

- **High Administrative Overhead:** Front-desk staff spend significant time manually negotiating time slots, confirming bookings, and handling cancellations via telephone or in-person visits.
- **Inconvenient Scheduling Windows:** Clients are strictly restricted to placing booking requests during physical business hours.
- **Unplanned No-Shows & Double-Booking:** Lack of automated reminders leads to high no-show rates, while non-synchronized scheduling causes double-booking conflicts.

4. Users & Stakeholders Matrix

The system caters to distinct primary and secondary user groups, each serving specific operational roles within the scheduling ecosystem:

STAKEHOLDER ROLE	DESCRIPTION	OPERATIONAL FOCUS
User 1: Client / Customer	External end-users seeking to reserve services online.	Self-service discovery, real-time booking, schedule management.
User 2: Service Provider	Internal professionals (e.g., Doctors, Advisors, Consultants).	Slot management, schedule tracking, client interactions.
User 3: Administrator	Technical support personnel managing platform governance.	User management, system configuration, audit logging.
User 4: Front-Desk Staff	Operational desk personnel assisting physical site workflow.	Walk-in registrations, manual slot overrides, check-ins.

5. User-Wise Requirements & Feature Breakdown

User 1: Client / Customer Requirements

- **Account & Profile Management:** Self-registration, secure authentication, password recovery, and personal profile updates.
- **Service Discovery & Filtering:** Browse service catalog by categories, ratings, service provider profiles, or calendar date.
- **Appointment Management:** View interactive calendar with real-time open time slots, perform single-click bookings, and request rescheduling or cancellation within allowed policies.
- **Notifications & History:** Receive instant booking confirmation notifications, automated SMS/email reminders, and access past/upcoming appointment history logs.

User 2: Service Provider Requirements

- **Availability & Working Hours Configuration:** Define weekly shift schedules, daily break windows, slot durations (e.g., 30 mins), and holiday blockouts.
- **Provider Dashboard:** View daily, weekly, and monthly appointment rosters in list or calendar view. Manage appointment states (**Approved** , **Completed** , **No-Show**).
- **Client Context & Notes:** Review basic client history prior to sessions and attach consultation summary notes post-appointment.

User 3: System Administrator Requirements

- **User & Role Management:** Create, suspend, or update accounts for Clients, Service Providers, and Administrative Staff with strict role permissions.
- **System Configuration:** Maintain service categories, system-wide operation parameters, and